



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)

UNIT TEST

13-12-2020

JEE(Main+Advanced) : NURTURE COURSE (PHASE : TEPS)

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	C	A	B	A	B	C,D	A,C,D	B,C	A,B,C
SECTION-II	Q.	11	12								
	A.	B	A,B,C,D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	8.00	0.40 to 0.45	100.97	27.10 to 27.20	4.75 to 4.80	4.05 to 4.15				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	D	B	B	A	A,B	A,B,C,D	A,B,C,D	C,D
SECTION-II	Q.	11	12								
	A.	A,B	B,C,D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	22.00	3.00	300.00	10.00	5.00	6.00				

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	D	C	B	B	A,B,D	B,C,D	A,D	A,B
SECTION-II	Q.	11	12								
	A.	A,B	A,D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	50.00	9.28 or 9.29	5.00	60.00	3.00	1.00				



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PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(B)

2. Ans. (D)

3. Ans.(D)

Mole ratio of C : H : O is 1 : 2 : 1

So empirical formula is CH_2O

$$m = \frac{\Delta T_b}{K_b} \Rightarrow \frac{5.7 \times 1000}{M \times 94.3} = \frac{0.17}{0.51}$$

$$M \approx 180$$

$$\text{So, } (\text{CH}_2\text{O})_n = 180 \Rightarrow n = 6$$

$$\text{Molecular formula} = \text{C}_6\text{H}_{12}\text{O}_6$$

4. Ans.(B)

5. Ans. (B)

6. Ans.(A)

$$a - 2r = 67.0 \text{ pm}$$

$$\text{for bcc } 4r = \sqrt{3}a$$

$$\text{So, } a = 500 \text{ pm}$$

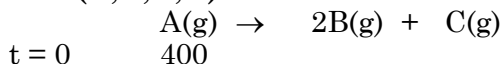
$$d = \frac{Z \times M}{N_A \times a^3}$$

$$d = 2.00 \text{ gm/cm}^3$$

7. Ans.(A,B)

8. Ans. (A,B,C,D)

9. Ans.(A,B,C,D)



$$t = 0 \quad 400$$

$$t = 20 \text{ min. } 400 - x \quad 2x \quad x$$

$$400 - x + 2x + x = 1000$$

$$x = 300 \text{ mm of Hg}$$

$$K = \frac{1}{20} \ln \frac{400}{100} = 0.693 \text{ min}^{-1}$$

$$t_{1/2} = \frac{\ln 2}{K} = \frac{0.693}{0.0693} = 10 \text{ min}^{-1}$$

$$\frac{30}{10} \times \ln 2 = \ln \frac{400}{400 - x}$$

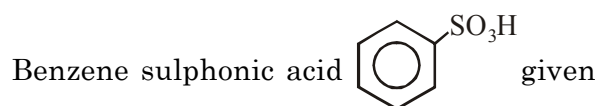
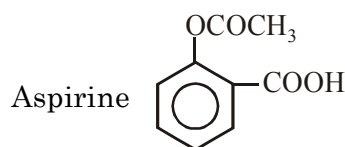
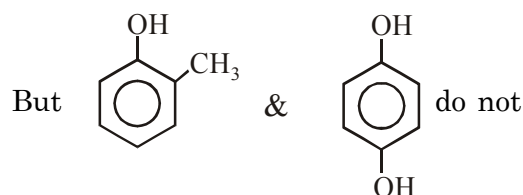
$$x = 350 \text{ mm of Hg}$$

$$P_c \text{ after 30 min} = 350 \text{ mm of Hg}$$

$$P_T \text{ after 30 min} = 1100 \text{ mm of Hg}$$

10. Ans.(C,D)

11. Ans. (A,B)

NaHCO₃ test

12. Ans.(B,C,D)

SECTION-II

1. Ans.(22.00)

2. Ans. (3.00)

3. Ans.(300.00)

Let n_B = mole of B present in 1 mole of mixture that is vaporized

$$y_B = \frac{n_B}{1}$$

mole fraction of B in liquid

$$\text{remaining} = x_B = \frac{1 - n_B}{1}$$

$$\text{As we know } P = P_A^0 + (P_B^0 - P_A^0)x_B \dots \dots \dots (1)$$

$$y_B = \frac{P_B^0 x_B}{P} \dots \dots \dots (2)$$

from eq. (1) & (2)

$$P = \sqrt{P_B^0 P_A^0} = \sqrt{100 \times 900} = 300 \text{ torr}$$

4. Ans.(10.00)

5. Ans.(5.00)

6. Ans.(6.00)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (C)**

(1, 2) & (5, 5) are foci of hyperbola so

$$2ae = 5$$

$$2a = 3$$

$$ae = 5/2$$

$$a = 3/2$$

$$\frac{3}{2}e = \frac{5}{2}, e = \frac{5}{3}$$

and we know that $\frac{1}{e^2} + \frac{1}{e'^2} = 1$

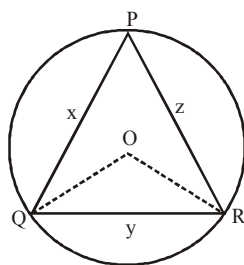
$$\frac{1}{e'^2} = 1 - \frac{9}{25} = \frac{16}{25}$$

$$\frac{1}{e'} = \frac{4}{5}$$

$$e' = \frac{5}{4}$$

$$8e' = 8 \times \frac{5}{4} = 10$$

2. **Ans. (A)**



$$x^2 + y^2 + z^2 = 5 \quad \angle QPR = 30^\circ$$

$$\frac{1}{2}xz \cdot \frac{1}{2} = \frac{\sqrt{3}}{4} \Rightarrow xz = \sqrt{3}$$

$$\cos \angle QPR = \frac{x^2 + z^2 - y^2}{2xz}$$

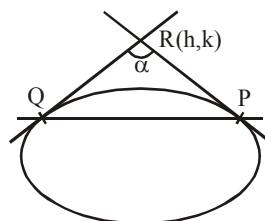
$$\frac{\sqrt{3}}{2} = \frac{5 - 2y^2}{2 \times \sqrt{3}} \Rightarrow 2y^2 = 2 \Rightarrow y = 1$$

from sine rule

$$= \frac{y}{\sin 30^\circ} = 2R \Rightarrow R = 1$$

3. **Ans. (D)**

Sol.



Chord of contact of R

$$hx + 2ky = 6$$

$$\text{compare with } \frac{x}{2} \cos \theta + y \sin \theta = 1$$

$$\frac{2h}{\cos \theta} = \frac{2k}{\sin \theta} = 6$$

$$h^2 + k^2 = 9$$

which is director circle of $x^2 + 2y^2 = 6$.

4. **Ans. (C)**

Sol. The image of A in $y = x$ will lie on BC.

So, $A' \equiv (6, 2)$

Now, BC is

$$y - 1 = \frac{1}{5}(x - 1) \Rightarrow x - 5y + 4 = 0$$

$$\Rightarrow \text{Ans} = \frac{4}{\sqrt{26}}$$

5. **Ans. (B)**

$$A = \cos \theta \cdot \cos 2\theta \cdot \cos 3\theta \dots \cos(n-1)\theta$$

$$B = \sin \theta \cdot \sin 2\theta \cdot \sin 3\theta \dots \sin(n-1)\theta$$

$$\text{where } \theta = \frac{2\pi}{2019}$$

$$n = 2019$$

$$2^{1009} \times A \times B = (2\sin \theta \cdot \cos \theta) (2\sin 2\theta \cdot \cos 2\theta)$$

$$\dots (2\sin(n-1)\theta \cdot \cos(n-1)\theta)$$

$$= \sin 2\theta \cdot \sin 4\theta \cdot \sin 6\theta \dots \sin(2n-2)\theta$$

$$\Rightarrow 2^{1009} \times A \times B = B$$

$$A = \frac{1}{2^{1009}} = 2^{-1009}$$

6. **Ans. (B)**

$(1+2x)^{10}(1+3x)^{10}$; for $(1+2x)^{10}$ numerically greatest term is T_8

$(1+3x)^{10}$ N.G.T. is $T_9 \Rightarrow T_8 T_9$ is N.G.T.

7. **Ans. (A,B,D)**

8. Ans. (B,C,D)

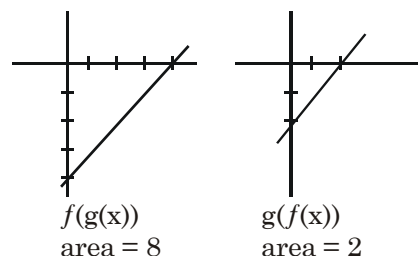
Sol. $f(x) = \begin{cases} -4-2x & x \in [-1,0) \\ -2x & x \in [0,1) \\ 4-2x & x \in [1,2) \\ 8-2x & x \in [2,3) \end{cases}$

$g(x) = \begin{cases} -2-\frac{x}{2} & x \in [-2,0) \\ -\frac{x}{2} & x \in [0,2) \\ 2-\frac{x}{2} & x \in [2,4) \\ 4-\frac{x}{2} & x \in [4,6) \end{cases}$

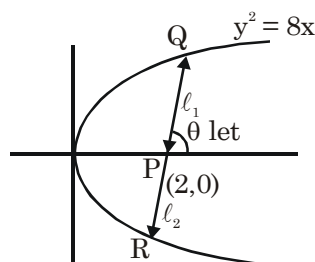
$f(x)$ and $g(x)$ both are bijective

$f(g(x)) = x - 4 \quad \forall x/I$

$g(f(x)) = x - 2 \quad \forall x/I$



9. Ans. (A,D)



Let equation of QR be

$$\frac{x-2}{\cos \theta} = \frac{y}{\sin \theta} = \ell$$

$$x = \ell \cos \theta + 2$$

$$y = \ell \sin \theta$$

if $(\ell \cos \theta + 2, \ell \sin \theta)$ lies on $y^2 = 8x$

$$\Rightarrow (\ell \sin \theta)^2 = 8(\ell \cos \theta + 2)$$

$$\Rightarrow \ell^2 \sin^2 \theta - 8 \cos \theta \cdot \ell - 16 = 0$$

here roots are ℓ_1 and $-\ell_2$

$$\Rightarrow \ell_1 - \ell_2 = \frac{8 \cos \theta}{\sin^2 \theta} \text{ and } -\ell_1 \ell_2 = -\frac{16}{\sin^2 \theta}$$

$$\text{Now harmonic mean} = \frac{2\ell_1 \ell_2}{\ell_1 + \ell_2}$$

$$= \frac{2\ell_1 \ell_2}{\sqrt{(\ell_1 - \ell_2)^2 + 4\ell_1 \ell_2}}$$

$$= 4$$

$$\text{we know that } G^2 = AH \Rightarrow G^2 = 4A$$

$$\Rightarrow G \propto \sqrt{A}$$

10. Ans. (A,B)

Sol. $P_1 : y^2 = \ell_1 x$

$$P_2 : y^2 = -\ell_2(x - b)$$

point on $P_1 \left(\frac{y^2}{\ell_1}, y \right)$ and $P_2 \left(b - \frac{y^2}{\ell_2}, y \right)$

find locus of midpoint of above points

11. Ans. (A,B)

Sol. $T_n = \begin{vmatrix} n-1 & n-2 & n-6 \\ 2n-4 & 2n-6 & 2n-11 \\ 3n-9 & 3n-12 & 3n-18 \end{vmatrix}$

$$C_2 \rightarrow C_2 - C_1, C_3 \rightarrow C_3 - C_2$$

$$T_n = \begin{vmatrix} n-1 & -1 & -4 \\ 2n-4 & -2 & -5 \\ 3n-9 & -3 & -6 \end{vmatrix}$$

$$R_2 \rightarrow R_2 - \frac{(R_1 + R_3)}{2}$$

$$T_n = \begin{vmatrix} n-1 & -1 & -4 \\ 1 & 0 & 0 \\ 3n-9 & -3 & -6 \end{vmatrix}$$

$$T_n = -1(-6) = 6$$

12. Ans. (A,D)

Sol. $\tan^{-1} x + \tan^{-1} \frac{1}{y} = \tan^{-1} 3$

$$x + \frac{1}{y} = 3$$

$$\Rightarrow xy + 1 = 3y - 3x$$

$$\Rightarrow x = \frac{3y-1}{y+3} = 3 - \frac{10}{y+3}$$

$$y+3 = 5, 10$$

$$y = 2, 7 \Rightarrow x = 1, 2$$

SECTION-II

1. Ans. 50.00

Sol. $\cos^{12}x + 3\cos^4x\sin^3x = \sin^9x - 1$

$$(\cos^4x)^3 + (-\sin^3x)^3 + 1^3 = 3\cos^4x(-\sin^3x)$$

$$\cos^4x + 1 = \sin^3x$$

$$\cos x = 0 \text{ and } \sin x = 1$$

$$x \in [0, 2\pi] \quad \text{No. of solution} = 1$$

$$x \in [0, 100\pi] \quad \text{No. of solution} = 50 = \lambda$$

2. Ans. 9.28 or 9.29

Sol. In $\triangle ABC$ $\Sigma \tan A = \Pi \tan A$

$$2 + \frac{3}{2} + \tan C = 2 \cdot \frac{3}{2} \tan C$$

$$\tan C = \frac{7}{4}, \quad \sin C = \frac{7}{\sqrt{65}}$$

$$R = \frac{c}{2\sin C} = \frac{2\sqrt{65}}{2 \cdot 7} \times \sqrt{65} = \frac{65}{7}$$

3. Ans. 5.00

$$(720 + 1)^{720}$$

$$\Rightarrow {}^{720}C_0 720^{720} + \dots + {}^{720}C_{719} \cdot 720 \cdot 1 + {}^{720}C_{720} \cdot 1^{720}$$

$$\Rightarrow \underbrace{{}^{720}C_0 \cdot 720^{720} + \dots}_{1000K} \underbrace{{}^{720}C_{720} \cdot 1^{720}}_{\dots 400} + 1$$

So we have to check last two terms ... 401

hence last 3 digits are 401 $\Rightarrow a + b + c = 5$

4. Ans. 60.00

$$1024 \times (225)^2 \times (25)^7 = 2^{10} \times 3^4 \times 5^{18}$$

2 & 5 are of the form $3k + 2$.

If two number of the form $3k + 2$ are multiplied together then we get a number of the form $3k + 1$

$\Rightarrow$ Divisor must contain even powers of 2 & 5 and it must not contain any power of 3.

$\Rightarrow$ Number of such divisors = $6 \times 10 = 60$ where number of ways to select power of 2 = 6

& number of ways to select power of 5 = 10

5. Ans. 3.00

Sol. $f\left(x + \frac{1}{2}\right) + f\left(x - \frac{1}{2}\right) = f(x) \quad \dots\dots(i)$

$$\text{put } x \rightarrow x + \frac{1}{2}$$

$$\Rightarrow f(x + 1) + f(x) = f\left(x + \frac{1}{2}\right) \quad \dots\dots(ii)$$

$$(i) + (ii) \text{ gives } f(x + 1) + f\left(x - \frac{1}{2}\right) = 0$$

again $x \rightarrow x + \frac{1}{2}$ gives period = 3

6. Ans. 1.00

